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Multiple aggregators and fairness criteria in the rolling-horizon OPF

Orchestration of the directional sweep, reduction of the computation time and the resulting mathematical formulation

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Reading convention. Everything that appears in red is new: it did not exist in the inherited formulation. The rest is reproduced unchanged and is included so that the model is explicit from beginning to end. Part 1 does not alter any equation — it is exclusively orchestration and performance, and its output is bit-identical to that of the previous model.

In blue is what is proposed and not yet implemented: it is included so that it can be discussed, and it is not part of any measured result in this document.

Unbalanced three-phase current-injection OPF model, in AIMMS with CPLEX. Case study: low-voltage feeder, transformer TR9.

1 Rolling-horizon orchestration and computation time

The rolling horizon runs through 48 half-hour periods. At each "now" period the day is rotated so that *now* is slot 1 and the **service block** $t^\star$ is slot 2; a committed dispatch is solved, it is executed, and 12 directions are swept to trace the region offered at $t^\star$. The formulation did not change: what changed is *what is solved and how many times*.

Both changes were found by profiling the existing sweep, and both are safe only because of how that sweep was already built: the 12 directions genuinely share one feasible region, and the un-rotated inputs are already frozen once per run. What changed is the placement of work that was reasonable when the sweep was written and no longer pays off in the configuration now used in production, a continuous QCP solved by barrier. Reviewed and agreed before pushing.

1.1 The loop, before and after

Before — 25 solves and 13 loads per period

```
for each period t:
 load DB + Excel
 solve MinImports                                 (1)
 execute slot 1, advance SOC
 for each direction θ_k, k = 1..12:
   load DB + Excel
   solve MinImportsMaxReverseP
   solve FOR_Vertex(θ_k)
```

Now — 13 solves and 1 load per period

```
load DB + Excel (once per run)
for each period t:
 light reset
 solve MinImports                                 (1)
 execute slot 1, advance SOC
 for each direction θ_k, k = 1..12:
   light reset
   solve FOR_Vertex(θ_k)
```

From one period to the next **one single thing** is carried over: the realised state of charge e_b^0, which advances with the executed dispatch of slot 1 and comes out of the *baseline*, not out of the sweep. The sweep is handed two: that e_b^0 and the dispatch of slot 1 *fixed*, which guarantees that the offered region is reachable from what the batteries actually do.

1.2 Where the time went

The removed solve, $\min\ \mathrm{OF}^{\text{imp+rev}}$, ran once per direction. Its objective **does not contain the angle**, and within a period the 12 directions share exactly the same feasible region: same reset, same data rotation, same e_b^0 and the same fixed slot 1. Therefore

> The 12 solves of a period were **the same optimization problem** — same matrix, same objective, same solution — solved 12 times. Its purpose was to seed the vertex solve: the answer was kept only as the starting point (*warm start*), and nothing else downstream read it.

1.3 The interventions

(E) Splitting the loader. The reading of the database and Excel was split into a *static* half, which runs once per run, and a per-solve *reset* that restores only the variables that the emptying destroys: the fixing of $q_{b,t}$, the flat voltage seed and the linearization point. Data loads per period: 13 → 1.

(A) Removal of the per-direction warm-start solve. It was suppressed. Solves per period: 25 → 13. It remains behind a switch that restores the previous behaviour, and it is kept *active* in the MIQCP validations, where the start channel is real.

(C) Fix in the diagnostic passes. On splitting the loader, the retries of failed directions stopped inheriting the experimental SOC floor; it is re-applied explicitly so that the retry runs under the same condition as the direction that failed.

1.4 Interior point: why the warm start did not help here

What a *warm start* is. Handing the solver a starting point already close to the answer, so that less ground is left for it to cover. It is standard practice and it usually helps.

Why it did not help here. It depends on the method, and the two available ones look for the optimum in opposite ways:

	Simplex	**Barrier (interior point)**
Where it advances	along the **edge**, jumping from vertex to vertex	through the **interior**, following the central path
Ideal starting point	a vertex close to the optimum — a basis	a **well-centred** point, far from every active constraint
Effect of starting on the boundary	it brings it closer: few jumps are missing	it **penalises** it: it must first re-centre

The point that the code handed over was the optimum of *another* problem, the minimum-cost one. And the optimum of a convex program with a linear objective is, by construction, an **extreme point**: stuck against the boundary, with several active constraints. It is exactly what an interior-point method does not want — hence the name of the method: there is a barrier term that pushes it *away* from the constraints, and starting on one of them forces that work to be undone before it can advance. In the corridor image: barrier walks down the middle, and it was being handed the point flattened against the wall.

The change. Suppress that solve. *No equation was touched.* The vertex solve now starts from the flat point that the reset leaves — all voltages at their balanced nominal value —, comfortably interior and with no active constraints, which is just what barrier needs.

Measured result: 1.27 s → 0.62 s per vertex solve. *Half, from no longer helping it.*

> **An honest nuance about the mechanism.** What is verified is the *measurement*: 2.05× per solve, with bit-identical output. The explanation of the badly centred point is the best supported one, but it has a known tension: on the continuous path the solver does not receive a basis — that channel exists only on the integer path — and the first solve, also a barrier one, does not produce one to hand over either. That the difference is measured all the same indicates that something of the previous state has an influence through a route that was not identified. The *what* is proven; the *why* is a supported hypothesis, not a verified fact.

Note that, even accepting the warm-start theory, **11 of the 12 solves were redundant anyway**: it was enough to solve it once per period and reuse the point. Both programs declare the same sets of constraints and variables, so that they generate a row-for-row identical matrix and differ only in the objective column — which is why removing one does not alter the other.

Numerical options kept: feasibility tolerance 10^{-3}, barrier convergence tolerance 10^{-2}, aggressive scaling, and the compact generator of quadratic programs — needed because the inherited generator expanded the quadratics and exhausted the memory towards period 37 of 48.

1.5 Measured results

	Before	Now	Factor
Micro network — TR9_two, 12 batteries, 48 periods × 12 directions			
Total time	41 m 22 s	10 m 12 s	4.06×
Vertex solves	731.0 s	358.7 s	2.05×
Per solve	1.27 s	0.62 s	
Production — TR9, 90 prosumers			
Total time	~28 h	**3 h 40 m**	~ **7.6**×
Vertex solves	30 213.4 s	8 440.7 s	3.58×
Per solve	52.45 s	14.65 s	

Equivalence verified. max $|\Delta\,\text{proj}| = 0.000\text{e}{+}00$ in all four runs; 576/576 optimal vertices; byte-identical state-of-charge file; the six metrics of the deliverable reproduced exactly. The acceptance criterion is the **projection** $\theta^\top(P, Q)$, not the equality of (P, Q): a flat face perpendicular to the sweep direction turns the argument of the maximum into a segment, and the returned point may legitimately slide along it.

The real bottleneck was not the data reading but the **generation of the mathematical program**: on TR9 each generation of a QCP with 90 batteries and 341 buses costs of the order of 30 s, and (A) removes 12 generations per period on top of 12 solves.

1.6 Second phase: one matrix generation per period

New. The first phase exposed the dominant remaining cost: **generating** the mathematical program. A `solve` statement in AIMMS does two things — it walks the algebraic model and expands it into explicit matrix rows, and only then calls the solver — and that generation happens on *every* `solve`, whether or not the data changed, because no matrix is cached between statements. On the whole feeder that expansion is 7.5 million rows and costs ~665 s, against ~200 s of actual solving.
The **GMP** library (*Generated Mathematical Program*) makes it possible to generate that matrix once and keep it in memory to work on: it can be modified and re-solved without being rebuilt. Since the 12 directions of a period **share one feasible region** and differ only in the objective, one generation can serve all 12 solves: per direction only the two coefficients of the row defining OF_{FOR} are rewritten.

Three interventions. (G) the sweep is served by a matrix generated once per period; (B) the committed dispatch is solved on *that same* instance by switching the objective column, which removes the second generation of the period; (F) the results file is closed after every vertex, so an interruption keeps what has already been solved.

Two details make (B) exact. The slot-1 dispatch moves from the `.nonvar` property — which belongs to the *model* and is read when the matrix is built, so setting it afterwards has no effect — to fixing the bounds of the *instance*. And the fairness rows, which the sweep must see and the committed dispatch must not, are **deactivated** for the dispatch solve and reactivated for the sweep: a deactivated row stays in the instance but is not passed to the solver, which is exactly what regenerating without it used to achieve. That is what lets one matrix per period serve *any* fairness level, not only the case with no criterion.

1.7 Results of the second phase

Production TR9. The three measured criteria, 48 periods × 12 directions each, with the state of charge carried forward between periods:

Criterion	max $|\Delta\,\text{proj}|$	Status	Result
$m = 0$ (no fairness)	1.0e−06	576/576 optimal	equivalent
$m = 1$ (prosumers)	1.0e−06	576/576 optimal	equivalent
$m = 2$ (aggregators)	**0.000e+00**	576/576 optimal	equivalent

The tolerance is 2e−06 and the file prints six decimals, so 1e−06 p.u. is *one* unit of the last printed digit: 1 W at $S_b = 1$ MW. In time, the three runs together go from **12 h 17 m** to **9 h 37 m**, a −22 %. Accumulated with the first phase, a production run of TR9 goes from ~28 h to ~3 h 10 m.

Whole feeder. This is the underlying question — whether the model can be run over the entire network — and it is where generation dominates, because it scales with problem size while the solver does not scale in the same proportion:

	Wall/vertex	**Solver**	**Reconstruction**
Without (G)	844–904 s	183–238 s	~665 s
With (G)	~**240 s**	236 s	~0

Wall-clock converges to solver time: the per-direction reconstruction disappears, a factor ~ 3.6×. That in turn makes the cost of the **exact** formulation on the whole feeder measurable, which is the one production uses:

			Simulated day (48 periods)	
Formulation	Solver/vertex	Generation	(G) only	(G)+(B)
Linearized	~236 s	~665 s	~2.4 days	~2.1 days
Exact	287.24 s	~1 005 s	~3.5 days	~2.9 days

The whole-feeder measurement was taken with (G) alone, which is what removes the 11 redundant reconstructions of the sweep. (B) additionally removes the generation of the committed dispatch — the only one left in the period — and is therefore worth a further **−15 % to −16 %** depending on the formulation. The last two columns are projections over 48 periods.

The exact formulation solves on the whole feeder. 12 of 12 directions optimal, no solve near the time limit, with true $V_{\max} = 1.10000$ p.u. and overvoltage violation at numerical zero. The matrix is 7 534 679 rows, 6 095 687 columns and 26 929 278 nonzeros. It costs +22 % per vertex over the linearized one and +43 % per period once generation is counted: **a fraction, not a multiple**.
This matters because the linearized variant relaxes the voltage limit — its tangent plane admits points the exact constraint rejects — and on the same period it **overstates the region by between 0.9 % and 1.4 %**, in the optimistic direction. The simulated-day figures are projections from representative periods, not complete runs.

As a practical consequence, the model's procedures can be launched **from the terminal** without opening the interface (`AimmsCmd.exe --run-only <Procedure> <project>.aimms`), which makes these long runs scriptable and observable from outside.

2 Formulation: multiple aggregators and fairness criteria

Unbalanced three-phase current-injection model, with linearization of the load and generation injections around a point $(v_0^{\text{re}}, v_0^{\text{im}})$ and exact quadratic constraints for maximum voltage and inverter capacity. Everything in red is new.

2.1 Sets and indices

$\mathcal{N}$	buses n	$\mathcal{B}$	batteries / prosumers b
$\mathcal{L}$	lines ℓ	$\mathcal{F} = \{1, 2, 3\}$	phases f, h
$\mathcal{K}$	transformers k	$\mathcal{H} = \{1, \ldots, 48\}$	periods t
$\mathcal{D}$	loads with PV d	$\mathcal{A}$	aggregators α

$t^\star$ denotes the service block; in the rolling horizon $t^\star = 2$ after the rotation. $a(b) \in \mathcal{A}$ is the aggregator of prosumer b, an input datum.

2.2 Objective function of the directional sweep

For each direction $\theta_k = 30°(k-1)$, $k = 1, \ldots, 12$:

$$\min \quad -\big(\cos\theta_k \, P_{t^\star} + \sin\theta_k \, Q_{t^\star}\big) \tag{1}$$

with the power at the connection point, positive from grid towards feeder,

$$P_t = \sum_g v^{\text{re}}_{\text{pcc},f(g)} \, i^{\text{re}}_{g,t} + v^{\text{im}}_{\text{pcc},f(g)} \, i^{\text{im}}_{g,t} \tag{2}$$

$$Q_t = \sum_g v^{\text{im}}_{\text{pcc},f(g)} \, i^{\text{re}}_{g,t} - v^{\text{re}}_{\text{pcc},f(g)} \, i^{\text{im}}_{g,t} \tag{3}$$

Minimising $-\theta_k^\top(P, Q)$ evaluates the *support function* of the feasible set in the direction θ_k; by rotating θ the boundary of the region is traced.

2.3 Network

The voltage is anchored at the reference bus r, fixed and balanced, and it is what sets the origin of the whole system:

$$v^{\text{re}}_{r,f,t} = \begin{cases} 1 & f = 1 \\ -\frac{1}{2} & f \in \{2,3\} \end{cases} \qquad v^{\text{im}}_{r,f,t} = \begin{cases} 0 & f = 1 \\ -\frac{\sqrt{3}}{2} & f = 2 \\ +\frac{\sqrt{3}}{2} & f = 3 \end{cases} \tag{4}$$

Current balance at each bus and phase:

$$\sum_{g \in n,f} i^{\text{re}}_{g,t} + \sum_{\ell:\,\text{to}(\ell)=n} i^{\text{re}}_{\ell,f,t} + \sum_{k:\,\text{to}(k)=n} i^{\text{re}}_{k,f,n,t} - \sum_{\ell:\,\text{from}(\ell)=n} i^{\text{re}}_{\ell,f,t} - \sum_{k:\,\text{from}(k)=n} i^{\text{re}}_{k,f,n,t} = \sum_{d \in n,f} \big(i^{\text{re,L}}_{d,t} - i^{\text{re,G}}_{d,t}\big) + \sum_{b \in n,f} i^{\text{re}}_{b,t} \tag{5}$$

and its analogue in the imaginary part. Voltage drop on lines, with mutual coupling:

$$v^{\text{re}}_{\text{from}(\ell),f,t} - v^{\text{re}}_{\text{to}(\ell),f,t} = \sum_{h \neq f} \big(R^m_\ell i^{\text{re}}_{\ell,h,t} - X^m_\ell i^{\text{im}}_{\ell,h,t}\big) + R^s_\ell i^{\text{re}}_{\ell,f,t} - X^s_\ell i^{\text{im}}_{\ell,f,t} \tag{6}$$

$$v^{\text{im}}_{\text{from}(\ell),f,t} - v^{\text{im}}_{\text{to}(\ell),f,t} = \sum_{h \neq f} \big(X^m_\ell i^{\text{re}}_{\ell,h,t} + R^m_\ell i^{\text{im}}_{\ell,h,t}\big) + X^s_\ell i^{\text{re}}_{\ell,f,t} + R^s_\ell i^{\text{im}}_{\ell,f,t} \tag{7}$$

Dy transformers with tap ratio $\tau_{k,t}$, coupling phase f with the next two:

$$v^{\text{re}}_{\text{from}(k),f,t} = -\frac{\nu_k \tau_{k,t}}{3}\Big(2v^{\text{re}}_{\text{to}(k),f\oplus 1,t} + v^{\text{re}}_{\text{to}(k),f\oplus 2,t} + R_k\big(2i^{\text{re}}_{k,f\oplus 1,\cdot,t} + i^{\text{re}}_{k,f\oplus 2,\cdot,t}\big) - X_k\big(2i^{\text{im}}_{k,f\oplus 1,\cdot,t} + i^{\text{im}}_{k,f\oplus 2,\cdot,t}\big)\Big) \tag{8}$$

The v^{im} equation is the same with R_k acting on the imaginary currents and $+X_k$ on the real ones. The primary–secondary current relation is $i^{\text{re}}_{k,f,\text{from},t}\,\nu_k\tau_{k,t} = i^{\text{re}}_{k,f,\text{to},t} - i^{\text{re}}_{k,f\oplus 1,\text{to},t}$, and analogous in i^{im}. In transformers without on-load tap changing the tap is fixed, $\tau_{k,t} = \tau^{\text{fix}}_k$, so that the product $\tau_{k,t}\,v$ introduces no nonlinearity: the model remains affine in the injections.

Voltage limits — upper exact, lower on the first-order approximation $\widehat{V^2}$ — and thermal limits:

$$(v^{\text{re}}_{n,f,t})^2 + (v^{\text{im}}_{n,f,t})^2 \le \overline{V}^2_n, \qquad \widehat{V^2}_{n,f,t} \ge \underline{V}^2_n \tag{9}$$

$$(i^{\text{re}}_{\ell,f,t})^2 + (i^{\text{im}}_{\ell,f,t})^2 \le \overline{I}^2_\ell, \qquad (i^{\text{re}}_{k,f,\text{from},t})^2 + (i^{\text{im}}_{k,f,\text{from},t})^2 \le \overline{I}^2_k \tag{10}$$

where $\widehat{V^2}_{n,f,t} = (v^{\text{re}}_0)^2 + (v^{\text{im}}_0)^2 + 2v^{\text{re}}_0(v^{\text{re}} - v^{\text{re}}_0) + 2v^{\text{im}}_0(v^{\text{im}} - v^{\text{im}}_0)$. The load and PV injections are linearized in the same way, around $(v^{\text{re}}_0, v^{\text{im}}_0)$:

$$i^{\text{re,L}}_{d,t} = \frac{P^0_{d,t}v^{\text{re}}_0 + Q^0_{d,t}v^{\text{im}}_0}{(v^{\text{re}}_0)^2 + (v^{\text{im}}_0)^2} + \frac{\partial i^{\text{re,L}}}{\partial v^{\text{re}}}\big(v^{\text{re}} - v^{\text{re}}_0\big) + \frac{\partial i^{\text{re,L}}}{\partial v^{\text{im}}}\big(v^{\text{im}} - v^{\text{im}}_0\big) \tag{11}$$

2.4 Batteries

With $p_{b,t}$ the net active power of prosumer b ($p_{b,t} > 0$ charging, $p_{b,t} < 0$ discharging) and $q_{b,t}$ the reactive power of the inverter:

$$p_{b,t} = p^{\text{ch}}_{b,t} - p^{\text{dis}}_{b,t}, \qquad p^{\text{ch}}_{b,t},\ p^{\text{dis}}_{b,t} \in [0, \overline{P}_b] \tag{12}$$

$$p_{b,t} = v^{\text{re}}_{0,b,t}\,i^{\text{re}}_{b,t} + v^{\text{im}}_{0,b,t}\,i^{\text{im}}_{b,t}, \qquad q_{b,t} = -v^{\text{re}}_{0,b,t}\,i^{\text{im}}_{b,t} + v^{\text{im}}_{0,b,t}\,i^{\text{re}}_{b,t} \tag{13}$$

$$e_{b,t} = e^0_b + \frac{\Delta t}{E_b}\sum_{\tau\le t}\Big(\eta^c_b p^{\text{ch}}_{b,\tau} - \frac{p^{\text{dis}}_{b,\tau}}{\eta^d_b}\Big), \qquad \underline{\sigma}_b \le e_{b,t} \le \overline{\sigma}_b \tag{14}$$

where $e_{b,t}$ is the state of charge as a fraction and $\underline{\sigma}_b, \overline{\sigma}_b$ its bounds — distinct from the capacity E_b, which appears in the denominator and is in units of energy. Inverter capacity at the service block (exact quadratic cone) and cyclic return of the state of charge over the window:

$$p^2_{b,t^\star} + q^2_{b,t^\star} \ \le\ \overline{P}^2_b, \qquad \Big|\sum_{t\in\mathcal{H}}\Big(\eta^c_b p^{\text{ch}}_{b,t} - \frac{p^{\text{dis}}_{b,t}}{\eta^d_b}\Big)\Big| \le \beta E_b/\Delta t \tag{15}$$

Reactive power is released only at $t^\star$: $q_{b,t} = 0$ for $t \ne t^\star$. In the sweep, the dispatch of slot 1 is fixed to the committed value: $p^{\text{ch}}_{b,1} = \hat{p}^c_b$, $p^{\text{dis}}_{b,1} = \hat{p}^d_b$. The model also keeps the inherited constraint $p_{b,t^\star} \le \kappa\,\overline{P}_b$, where $\kappa = 0$ reproduces the assumption of the paper — batteries only discharging at the service block — and $\kappa = 1$ is the production configuration, bidirectional; with $\kappa = 1$ the constraint is dominated by $p^{\text{ch}}_{b,t^\star} \le \overline{P}_b$ and is not binding.

2.5 Fairness: proportional criterion C1

> **New.** Let s_b be the nominal capacity of prosumer b — $s_b = \overline{P}_b$ is adopted, which in this model is at once the power of the battery and the radius of the circle of inverter capacity. Let $\zeta \in \mathbb{R}$ be a **single free scalar per solve**, interpretable as the *utilisation factor* of the fleet. With $m \in \{0, 1, 2\}$ the fairness level and $\chi \in \{0, 1\}$ the activation gate:

$$p_{b,t^\star} = \zeta\, s_b \qquad \forall b \in \mathcal{B}, \qquad \text{if } m = 1 \text{ and } \chi = 1 \tag{16}$$

$$\sum_{b:\, a(b)=\alpha} p_{b,t^\star} = \zeta \sum_{b:\, a(b)=\alpha} s_b \qquad \forall \alpha \in \mathcal{A}, \qquad \text{if } m = 2 \text{ and } \chi = 1 \tag{17}$$

A single degree of freedom. The criterion is written with *one* scalar ζ, which removes *one* degree of freedom. Writing it as $n-1$ pairwise equalities would remove $n-1$ and would overestimate the cost of fairness by a factor close to two.

Scope of each level. Level 2 constrains only the sums per aggregator; the internal split of each aggregator remains free. The levels are alternative, not cumulative.

Consequence of the sign. Since $p_{b,t^\star} = p^{\text{ch}} - p^{\text{dis}}$, a single ζ forces the whole fleet to the same sign: *it forbids one battery to charge while another discharges*. It is the strictest reading of the criterion, and that is why C1 gives the **upper bound** on the cost of fairness.

Feasibility. $\zeta = 0$ is always admissible, so that C1 *contracts* the region and never makes it infeasible.

Corrected (31 Jul). The bound on the scalar, and why it is not a constraint. The bound is stated as a *linear box*, never in modulus: writing $|\zeta| \le 1$ would introduce a nonlinearity for no reason. The correct form is

$$-1 \;\le\; \zeta \;\le\; 1. \tag{18}$$

That said, (18) **is not added to the model**, because it holds by construction. Every battery already satisfies $|p_{b,t^\star}| \le \overline{P}_b$ through its own charge and discharge limits, which are box bounds on p^{ch} and p^{dis} and hold at all times, independently of the fidelity switches. With $s_b = \overline{P}_b$, dividing (16) by the capacity carries that bound straight onto ζ; at level 2 the same follows from the triangle inequality on the aggregator sum:

$$\Big|\sum_{b:\, a(b)=\alpha} p_{b,t^\star}\Big| \;\le\; \sum_{b:\, a(b)=\alpha} |p_{b,t^\star}| \;\le\; \sum_{b:\, a(b)=\alpha} s_b.$$

Imposing it would therefore add a redundant row. The *a posteriori* verification shows further that the bound is **reached** —max $|\zeta| = 1.0000$— and never exceeded: it is an effective facet of the feasible region, not slack.

Warning for $m = 3$ and $m = 4$. The argument requires the normalising capacity to be *the same quantity* that bounds the left-hand side. When the criterion moves to the prosumer's export the normalisation is no longer $\overline{P}_b$ and the bound no longer comes for free: it is re-derived in §2.7, where it survives on one side only.

2.6 The gate χ

χ takes the value 1 in the 12 directional solves and 0 in the solve of the committed dispatch. The reason is not one of implementation:

> $t^\star$ is slot 2 *also* in the baseline, so that without the gate (16)–(17) would constrain the dispatch executed in slot 1, the trajectory of the state of charge would diverge from the case without fairness, and the comparison would measure two mixed effects: the contraction of the region, and a different battery state. Moreover the objective of the baseline contains no service term at $t^\star$ — by construction $w_2(t^\star) = 0$ — so imposing a market rule there would constrain an agent that obtains no value at all from participating in it.

2.7 Fairness on the exchange: $m = 3$ and $m = 4$

New. Levels 1 and 2 share out the *battery balance*. The review of 31-Jul-2026 proposed applying the same proportional criterion to the **prosumer's exchange with the network**, which is the quantity a flexibility market actually observes: consumption minus PV generation, plus charge and minus discharge.

Let x_b be the **net exchange of prosumer b with the network** at the service block, positive towards *export*. With $g_{b,t}$ and $c_{b,t}$ the prosumer's PV generation and consumption — both input data — and $p_{b,t}$ the battery balance already defined in §2.4:

$$x_b = \underbrace{g_{b,t^\star} - c_{b,t^\star}}_{\text{passive surplus}} - \underbrace{p_{b,t^\star}}_{= p^{\text{ch}}_{b,t^\star} - p^{\text{dis}}_{b,t^\star}} \tag{19}$$

Charging the battery ($p_{b,t^\star} > 0$) reduces what the prosumer delivers to the network and discharging increases it. The criterion is then written on x_b instead of on $p_{b,t^\star}$:

$$x_b = \zeta\, s_b \qquad \forall b \in \mathcal{B}, \qquad \text{if } m = 3 \text{ and } \chi = 1 \tag{20}$$

$$\sum_{b:\, a(b)=\alpha} x_b = \zeta \sum_{b:\, a(b)=\alpha} s_b \qquad \forall \alpha \in \mathcal{A}, \qquad \text{if } m = 4 \text{ and } \chi = 1 \tag{21}$$

With $m \geq 3$ the reference capacity is no longer just the battery nameplate but the export capability available at the service instant, $s_b = \overline{P}_b + g_{b,t^\star}$, which is the quantity homogeneous with x_b.

A consequence that is not a defect: the criterion can be infeasible. Under $m = 1$ the value $\zeta = 0$ parks the fleet and is always admissible, so the criterion only contracts the region. Under $m = 3$ that no longer holds: the passive net load enters as a **different constant for each prosumer**, so $\zeta = 0$ requires the battery to offset its own net consumption exactly, which it cannot always do.

Measured evidence (micro bed, 3-Aug-2026). In the 22:00 block all twelve directions come out infeasible: one battery at its state-of-charge floor, with no discharge headroom and 0.93 kW of its own consumption, imposes $\zeta \leq -0.1315$ on the *whole* fleet, and no single ζ satisfies the rest simultaneously. **It is a result about the criterion, not a numerical failure.**

The derived bound also changes shape. Under $m = 1$ and $m = 2$, $|\zeta| \leq 1$ holds by construction. Under $m = 3$ and $m = 4$ only one side survives: export is bounded by $g_{b,t^\star} + \overline{P}_b \leq s_b$, hence $\zeta \leq 1$; but import is load plus battery power, and a prosumer whose consumption exceeds its generation plus its battery can push ζ below -1. It is worth reporting the realised range rather than assuming it.

2.8 Proposal: fairness on the available range ($m = 5$ and $m = 6$)

Proposed, not yet implemented. Criteria $m = 1$ to $m = 4$ share out in proportion to what is *installed*. The problem is that the nameplate does not know what the prosumer can do at this instant: a full battery has full nameplate and zero import capability, and the equality asks it for something impossible — which is the origin of the infeasibility of $m = 3$.

The proposal shares out in proportion to the **exchange range each prosumer can actually traverse** at the service block. Since the slot-1 dispatch is already committed when the sweep runs, both ends of that range are *parameters*:

$$x_b \in [\,\underline{x}_b,\ \overline{x}_b\,], \qquad \underline{x}_b = -\big(n_b + \kappa_b\big), \qquad \overline{x}_b = \delta_b - n_b$$

$$x_b = \underline{x}_b + \zeta\,(\overline{x}_b - \underline{x}_b) \qquad \forall b \in \mathcal{B}, \qquad \text{if } m = 5 \text{ and } \chi = 1 \tag{22}$$

$$\sum_{b:\,a(b)=\alpha} x_b = \sum_{b:\,a(b)=\alpha} \underline{x}_b + \zeta \sum_{b:\,a(b)=\alpha} (\overline{x}_b - \underline{x}_b) \qquad \forall \alpha \in \mathcal{A}, \qquad \text{if } m = 6 \text{ and } \chi = 1 \tag{23}$$

with $n_b = c_{b,t^\star} - g_{b,t^\star}$ the net consumption, and κ_b and δ_b the charge and discharge headroom the state of charge allows in that block — all three are *parameters*, because the slot-1 dispatch is already committed when the sweep runs.

Now ζ is not the fraction of the nameplate but the **relative position within each prosumer's own feasible range**, common to the whole fleet. Under $m = 6$ the aggregator must reach that same position *in aggregate*, while the split among its own prosumers stays free — the same relation $m = 2$ has with $m = 1$.

Why it closes the gap. For any $\zeta \in [0, 1]$ the resulting x_b falls inside the interval by construction, so it is **never infeasible** and no block is lost. It remains an **equality for everyone**, with no one-sided rows and no exceptions that apply only when convenient. It is **linear**: the ends are parameters and ζ multiplies a constant, with no binaries. The bound $0 \le \zeta \le 1$ is again **derived** rather than imposed. And the exclusion of whoever cannot participate **emerges from the formulation**: if a prosumer cannot move, $\overline{x}_b = \underline{x}_b$ and its row reads $x_b = \underline{x}_b$ whatever ζ is, so it neither constrains nor is constrained.
It is a different definition of fairness, not a correction. $m = 1$ to $m = 4$ share out in proportion to what is installed; $m = 5$ and $m = 6$, to what is available. That is why they are proposed as *new* criteria and do not replace the earlier ones: with the three families implemented the comparison can be shown measured rather than argued.

The alternative route — patching $m = 3$ — was explored and set aside. Excluding whoever cannot participate resolves the hard infeasibility but does not cover whoever can do little; relaxing the row to a one-sided inequality binds from below but does not prevent exporting beyond one's share, so the criterion would end up applying only when convenient; and making it depend on the sign of ζ — which is an outcome of the solve — would require a disjunction, that is binaries and a mixed-integer problem. The answer is a different normalisation, not a better patch.

2.9 Cases and nesting

m	Case	Constraint active at $t^\star$
0	No fairness (reference)	— no row generated
1	Proportional between prosumers	(16)
2	Proportional between aggregators	(17)
3	On the exchange, between prosumers	(20)
4	On the exchange, between aggregators	(21)
5	On the available range, between prosumers	(22)
6	On the available range, between aggregators	(23)

With $m = 0$ both domains are empty: no row is generated and ζ remains as a free column with no coefficients, which presolve removes. That is the reason why the reference case reproduces the previous model exactly.

Nesting. Level 1 implies level 2: by summing (16) over the prosumers of an aggregator, (17) is obtained with the same ζ. Therefore

$$\mathcal{X}(m{=}1) \subseteq \mathcal{X}(m{=}2) \subseteq \mathcal{X}(m{=}0) \quad \Longrightarrow \quad \mathrm{proj}_\theta^{(1)} \le \mathrm{proj}_\theta^{(2)} \le \mathrm{proj}_\theta^{(0)} \quad \forall \theta,\ \forall t \tag{24}$$

which is the acceptance test: a constraint cannot enlarge the feasible set, and any violation is a defect and not a result.

The nesting holds within each family. Levels 1–2 share out the battery balance and levels 3–4 the prosumer's exchange: they are different quantities, so $m = 3$ is not a special case of $m = 1$ nor the other way round, and their regions are not comparable by inclusion. Within the exchange family, summing (20) over the prosumers of an aggregator does return (21) with the same ζ, so $\mathcal{X}(m{=}3) \subseteq \mathcal{X}(m{=}4)$ as before.

2.10 Results on the micro network ($m = 1$, $m = 2$)

Full verification on the 576 vertices: the reference case reproduces the previous model with exact $\Delta = 0$ and byte-identical state of charge; the nesting holds without exception; and the constraint is satisfied with a maximum residual of 10^{-4}, an order stricter than the solver's own feasibility tolerance.

	$m = 1$ **prosumers**	$m = 2$ **aggregators**
Vertices with contraction	297 / 576 (51.6 %)	272 / 576 (47.2 %)
Maximum projection loss	38.3 kW	16.1 kW
Area loss — median	22.2 %	7.5 %
Area loss — maximum	**87.3 %**	23.1 %
Observed max $\lvert\zeta\rvert$ (bound (18), reached)	1.000	1.000

Two observations. *(i)* The loss of *area* is much larger than that of projection per direction, whose median is practically nil: where the boundary has a flat face perpendicular to the sweep direction, the optimum slides to a fair point at no cost in that direction. Both metrics must be reported. *(ii)* Scaling a two-dimensional region by γ scales its area by γ^2: an area loss of 87 % is equivalent to a linear shrinkage to ~36 %, not to 13 %.

The fleet is homogeneous — $\overline{P}_b = 3.6$ kW in the 1,216 batteries — so that on this data the proportional criterion coincides numerically with the egalitarian one. That does not make it innocuous: the network is radial, and two prosumers with the same battery at different electrical positions cannot inject the same amount without producing different voltage rises. The heterogeneity that makes C1 bind is **locational**, not equipment.

2.11 Results on TR9 in production

The same metrics on the production case — TR9, 90 prosumers, 48 periods × 12 directions — measured against the $m = 0$ reference case of the same run:

	$m = 1$ **prosumers**	$m = 2$ **aggregators**
Vertices with contraction	554 / 576 (96.2 %)	476 / 576 (82.6 %)
Maximum projection loss	315.9 kW	23.6 kW
Area loss — median	35.8 %	2.3 %
Area loss — maximum	90.1 %	6.8 %
Observed max $\lvert\zeta\rvert$	1.000	1.000
Non-optimal vertices	0 / 576	0 / 576

The structure repeats at production scale and sharpens: level 1 contracts almost every vertex and costs a median area loss of 35.8 %, whereas level 2 — which only ties the per-aggregator sums — stays at 2.3 %. The bound $\lvert\zeta\rvert \le 1$ is reached and never exceeded in either, as on the micro network.

2.12 Results with the criterion on the exchange ($m = 3$, $m = 4$)

On the micro bed, against the same reference case:

	$m = 3$ **prosumers**	$m = 4$ **aggregators**
Vertices with contraction	564 / 564 (100 %)	560 / 576 (97.2 %)
Maximum projection loss	39.7 kW	18.2 kW
Area loss — median	38.7 %	15.5 %
Area loss — maximum	95.5 %	35.0 %
Observed ζ — range	$[-1.014,\ 0.865]$	$[-1.038,\ 0.938]$
Infeasible vertices	**12 / 576**	0 / 576

The two warnings of §2.7, measured. The 12 infeasible vertices of $m = 3$ are *the twelve directions of a single block*, the 22:00 one — not scattered failures, but a whole block the criterion leaves without a region. And the bound stops being symmetric: $\zeta \leq 1$ holds with room to spare in both, but ζ drops to -1.014 and -1.038, below -1. This is exactly what the argument anticipates — export is bounded by s_b, import is not — and it is why the realised range is worth reporting rather than assuming.

3 Next steps

1. **Whole-feeder run, without fairness.** This is the validation still missing: a complete day of 48 periods over the entire network, with $m = 0$, to replace the ~2.9-day projections of §1.7 with measurement and to confirm optimality and results at that scale. No 48-period run on the whole feeder has been completed so far; what has been measured are representative periods. It is also worth carrying the incremental flush (F) over to the production sweep beforehand, since its results file is currently closed only at the end and an interruption of a multi-day run would leave nothing.

2. **Implement $m = 5$ and $m = 6$.** The formulation of §2.9 is defined but not built. With the three families available — proportional to the nameplate ($m = 1, 2$), on the exchange ($m = 3, 4$) and on the available range ($m = 5, 6$) — the comparison between definitions of fairness can be shown measured rather than argued, which is what allows it to be decided on evidence rather than preference.